

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2-Industry Problem Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4- Industry Problem Solving	Assessment:	Assessment Tool 2-Industry Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	<u>B.Purna Kishor Reddy</u>	Reg. No.:	192472171
Section / Batch:	2024/Slot-D	Date:	

Code of Conduct

I B.Purna Kishor Reddy (192472171) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.Purna Kishor Reddy

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly	

Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Hypothesis Testing Using Confidence Interval and P-Value

Approach

The following solutions apply the standard 10-step hypothesis testing framework — identifying the population and sample, defining the parameter of interest, computing the point estimate, stating H_0 and H_1 , selecting the significance level, calculating the confidence interval and/or test statistic, calculating the p-value, comparing it with alpha, and arriving at an industry-oriented conclusion.

1. Digital Marketing — Hypothesis Testing Using CI

Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

Solution

1. Population and Sample: Population: all website visitors exposed to the advertisement campaign.

Sample: $n = 1,000$ visitors who viewed the new advertisement.

2. Parameter of Interest: The true population proportion of visitors who convert after the campaign, denoted p .

3. Point Estimate: $\hat{p} = x/n = 90/1000 = 0.09$ (9%).

4. Null Hypothesis (H_0): $p = 0.08$ (the campaign has not improved the conversion rate beyond the baseline of 8%).

5. Alternative Hypothesis (H_1): $p > 0.08$ (the campaign has improved the conversion rate).

6. Significance Level (α): $\alpha = 0.05$ (95% confidence).

7. Confidence Interval / Test Statistic: Standard error: $SE = \sqrt{[\hat{p} (1-\hat{p})/n]} = \sqrt{[(0.09 \times 0.91)/1000]}$
=

$\sqrt{0.0000819} \approx 0.00905$. 95% CI = $\hat{p} \pm 1.96 \times SE = 0.09 \pm 1.96(0.00905) = 0.09 \pm 0.0177 = (0.0723, 0.1077)$, i.e. (7.23%, 10.77%).

8. P-value (equivalent check): $z = (0.09 - 0.08)/\sqrt{[(0.08 \times 0.92)/1000]} = 0.01/0.00858 \approx 1.17$;
onetailed p-value = $P(Z > 1.17) \approx 0.121$.

9. Compare with α : The baseline value 0.08 lies inside the 95% CI (0.0723, 0.1077), and the p-value (0.121) is greater than α (0.05).

10. Industry-Oriented Conclusion: There is not enough statistical evidence at the 95% confidence level to conclude that the new advertisement has improved the conversion rate. Although the observed rate (9%) is numerically higher than the baseline (8%), the increase could plausibly be due to random sampling variation. The marketing team should continue testing with a larger sample or run the campaign longer before rolling it out broadly.

2. Education Technology — Hypothesis Testing Using CI

An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students obtains an average score of 78 with standard deviation

10. Use a 95% confidence interval to assess the claim.

Solution

1. Population and Sample: Population: all students using the adaptive-learning platform.

Sample: $n = 64$ students.

2. Parameter of Interest: The true population mean test score, μ , of students using the platform.

3. Point Estimate: $\bar{x} = 78$, with sample standard deviation $s = 10$.

4. Null Hypothesis (H_0): $\mu = 75$ (the platform raises the average score to exactly the claimed minimum of 75).

5. Alternative Hypothesis (H_1): $\mu > 75$ (the platform's true average score exceeds 75, supporting or exceeding the claim).

6. Significance Level (α): $\alpha = 0.05$ (95% confidence).

7. Confidence Interval / Test Statistic: Standard error: $SE = s/\sqrt{n} = 10/\sqrt{64} = 10/8 = 1.25$. 95% CI $= \bar{x} \pm 1.96 \times SE = 78 \pm 1.96(1.25) = 78 \pm 2.45 = (75.55, 80.45)$.

8. P-value (equivalent check): $z = (78 - 75)/1.25 = 2.40$; one-tailed p-value $= P(Z > 2.40) \approx 0.0082$.

9. Compare with α : The claimed value of 75 lies below the entire 95% CI (75.55, 80.45), and the p-value (0.0082) is less than α (0.05).

- 10. Industry-Oriented Conclusion:** There is sufficient statistical evidence at the 95% confidence level to support the EdTech company's claim — the platform improves average test scores to at least 75, and the true mean is likely even higher (between roughly 75.55 and 80.45). This CI strengthens the company's marketing claim and can be cited in product validation reports.

3. Automobile Industry — Hypothesis Testing Using P-value

A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars gives a mean mileage of 19.2 km/litre with standard deviation 2.5 km/litre.

Perform a hypothesis test at the 5% significance level using the p-value approach. **Solution**

- 1. Population and Sample:** Population: all cars of the new model produced by the manufacturer.

Sample: $n = 50$ cars tested for mileage.

- 2. Parameter of Interest:** The true population mean mileage, μ , of the new car model (km/litre).
- 3. Point Estimate:** $\bar{x} = 19.2$ km/litre, with sample standard deviation $s = 2.5$ km/litre.
- 4. Null Hypothesis (H_0):** $\mu = 20$ km/litre (the manufacturer's claim is true).
- 5. Alternative Hypothesis (H_1):** $\mu \neq 20$ km/litre (the true average mileage differs from the claimed value — two-tailed test).
- 6. Significance Level (α):** $\alpha = 0.05$.
- 7. Test Statistic:** Standard error: $SE = s/\sqrt{n} = 2.5/\sqrt{50} = 2.5/7.071 \approx 0.3536$. $z = (\bar{x} - \mu_0)/SE = (19.2 - 20)/0.3536 = -0.8/0.3536 \approx -2.263$.
- 8. P-value:** Two-tailed p-value $= 2 \times P(Z < -2.263) \approx 2 \times 0.0118 = 0.0236$.
- 9. Compare with α :** p-value (0.0236) $< \alpha$ (0.05), so the result is statistically significant.
- 10. Industry-Oriented Conclusion:** There is sufficient evidence to reject H_0 . The new model's actual average mileage (19.2 km/litre) is significantly lower than the claimed 20 km/litre. The manufacturer should review the mileage claim used in marketing and specifications, as it may not hold up under quality-testing/regulatory scrutiny, or should investigate manufacturing/design factors causing the shortfall.

4. Cloud Computing — Hypothesis Testing Using P-value

A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and standard deviation of 12 ms. Test the claim at $\alpha = 0.05$ using the p-value method.

Solution

1. **Population and Sample:** Population: all server requests handled by the cloud provider.

Sample: $n = 81$ requests.

2. **Parameter of Interest:** The true population mean server response time, μ (in milliseconds).

3. **Point Estimate:** $\bar{x} = 104$ ms, with sample standard deviation $s = 12$ ms.

4. **Null Hypothesis (H0):** $\mu = 100$ ms (the provider's claimed average response time is correct).

5. **Alternative Hypothesis (H1):** $\mu \neq 100$ ms (the true average response time differs from the claim — two-tailed test).

6. **Significance Level (α):** $\alpha = 0.05$.

7. **Test Statistic:** Standard error: $SE = s/\sqrt{n} = 12/\sqrt{81} = 12/9 = 1.333$. $z = (\bar{x} - \mu_0)/SE = (104 - 100)/1.333 \approx 3.00$.

8. **P-value:** Two-tailed p-value = $2 \times P(Z > 3.00) \approx 2 \times 0.00135 = 0.0027$.

9. **Compare with α :** p-value (0.0027) < α (0.05), so the result is statistically significant.

10. **Industry-Oriented Conclusion:** There is sufficient evidence to reject H0. The server's actual average response time (104 ms) is significantly higher than the claimed 100 ms. This indicates a real performance gap, not random variation. The cloud provider should investigate infrastructure bottlenecks (network latency, server load, or scaling issues) and consider revising its published SLA (Service Level Agreement) response-time commitment.

5. E-commerce — Hypothesis Testing Using P-value

An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10% using the p-value approach at the 5% significance level.

Solution

1. **Population and Sample:** Population: all orders placed with the online company. Sample: $n = 1,000$ orders.
2. **Parameter of Interest:** The true population proportion of orders returned, p .
3. **Point Estimate:** $\hat{p} = x/n = 130/1000 = 0.13$ (13%).
4. **Null Hypothesis (H0):** $p = 0.10$ (the company's claimed return rate of 10% is correct).
5. **Alternative Hypothesis (H1):** $p > 0.10$ (the actual return rate is higher than claimed — onetailed test).
6. **Significance Level (α):** $\alpha = 0.05$.
7. **Test Statistic:** Standard error under H0: $SE = \sqrt{p_0(1-p_0)/n} = \sqrt{[(0.10 \times 0.90)/1000]} = \sqrt{0.00009} \approx 0.009487$. $z = (\hat{p} - p_0)/SE = (0.13 - 0.10)/0.009487 \approx 3.162$.
8. **P-value:** One-tailed p-value = $P(Z > 3.162) \approx 0.00078$.
9. **Compare with α :** p-value (0.00078) $< \alpha$ (0.05), so the result is statistically significant.
10. **Industry-Oriented Conclusion:** There is sufficient evidence to reject H0. The actual product return rate (13%) is significantly higher than the claimed 10%. This suggests underlying issues such as product quality, inaccurate listings, or sizing/fit problems. The e-commerce company should investigate root causes of returns and update its reported return-rate metric for stakeholders and investors.